

Dominic Morris

Chief Technology Officer, Principal Engineer, Agentic Architect

dom@d0m1.com // **Telegram** @dom_1976 // **WhatsApp** +65 9272 4977
projects.d0m1.com // github.com/7-of-9 // 7-of-9.github.io/tokens

Nationality British (Resident UK & Thailand) // **Availability** 2-4W

Full Stack Founder & Engineer: 20+ years' experience from startups to IPOs

- Agentic engineering (2025–present): production software built as a dark factory — Shapiro Level 5 (no manual human reads, writes or approvals): Wagey.gg - manually orchestrated, 14–15 concurrent LLM sessions run by hand. Heimdall Global - Yegge Stage 8 orchestration (agents orchestrating agents; the whole workflow from frozen spec to verified, merged code automated) over multi-vendor fleet, run rate half a million lines and 44B tokens a month (www.d0m1.com/tokens).
- Regulator facing: technology lead — CTO, CIO and CISO: I design and build out regulator-ready technology policy stacks (technology governance, infosec, outsourcing, BCP/DR, incident response) and own the regulator relationship; licensed venues built: [ACX](#)'s carbon market under ADGM FSRA (Recognised Investment Exchange and Recognised Clearing House), [SDAX](#)'s tokenized equity platform to MAS Recognised Market Operator, ayondo's FCA/BaFin real-money platform and its SGX IPO, [Heimdall's](#) FSRA application; front-office banking at Standard Chartered, Citi, RBS, Deutsche Bank and Barclays — FX options, equity and energy derivatives, real-time risk, market data, trade processing, order management; TradeHero's 28-exchange feeds and broker integrations.
- Lifelong builder: C/C++ (8086, Win32, 20+ years), C# (.NET Core/CLR, Unsafe context), JS/TS at significant LOC/complexity, Rust (Anchor), Solidity, basic Python; core background - low-level optimization, memory and data models, CPU, concurrency in high-throughput environments. UX from DOS thru to WinForms and React/Vite, Web3 integrations, DirectX (Managed C++ for Volatility Surface rendering & video game prototypes), WinForms; native iOS & Android apps; deep experience in specifying, detailing, designing, testing & iterating as a founder and end-to-end product owner. Relational data modelling in detail and at scale across SQL Server (deep T-SQL optimization experience), NoSQL Azure Table Storage & Redis.
- Blockchain: crypto native - building, testing & maintaining Solana/Rust (Anchor, PDA management), Solidity (Hardhat, Truffle), Web3.js, Geth (Go-Ethereum internals), WASM (C++ EOS), Trezor Blockbook, bitcoinjs-lib, bitgo-utxo-lib, and ERC-20/721/1400 smart contracts, since 2017.
- Process & Cloud: AWS (ECS Fargate, Cognito, Lambda), Azure (App Services, Functions, Service Fabric, Table Storage), Kubernetes with Argo CD GitOps, Docker; designing for auto-scaling, failover and 24/7 institutional availability. CI/CD (GitHub Actions, Azure DevOps, bespoke pipelines), policy-as-code gates, supply-chain security (signed images, SBOMs, vulnerability and secret scanning), observability (OpenTelemetry — metrics, logs, traces), test-domain modelling and automated regression for regulated and on-chain immutable deployments.

Commercial Experience

Heimdall Global // Fractional Chief Technology Officer

Aug 2026 – current www.heimdall.global

CTO of Heimdall Global, a pre-licence financial-markets-infrastructure venture (ADGM / FSRA). I architected a compliance-first regulated trading venue — deterministic matching engine, double-entry money ledger, tamper-evident audit trail, workflow-driven operations — and am building it as a dark factory (Shapiro Level 5: no human reads, writes or approves the code; verification moved up the stack into harness, context and verification engineering) with Yegge Stage 8 orchestration (agents orchestrating agents: the entire workflow from frozen spec to verified, merged code is automated across five harnesses and four model providers). I own intent, law, setpoints and the supervisory loop; agents write, test and verify every line.

Monthly agentic run rate:

- Code and delivery: 440k lines of code and configuration across 2,000 files (125k product — Go 76k, Rust 26k, SQL 12k — and 157k test); +1.33M / -200k lines churned; 3,700 commits, 870 merged PRs, 4,900 CI runs, 0 PR approvals, 0 change requests.
- Tokens and verification: 44B tokens (mean 1.48B a day, peak 4.97B, 96% cache reads), 150k model calls; 140 cold-verifier verdicts, 360 red-proofed tests, 92.9% of pushes to main green; steered by about 300 messages, 30 rulings and 3 gate signatures from me.
- Spec-Driven Delegation: 977-requirement PRD in 14 families, each requirement carrying executable acceptance criteria and a regulatory hook (168 rulebook citations), traced to 82 work packages with acceptance frozen per round.
- Hierarchical Multi-Agent Orchestration: One orchestrator seat dispatching builder lanes, cold verifiers, fix loops and gate pre-checks into isolated git worktrees across 5 harnesses, 4 providers and 15 models on 3 tiers; the seat re-seated across vendors under one constitution.
- Verification Design: A fresh-context cold verifier per round with its own mutants; red-proofed tests; 77 bespoke CI gates and 9 required checks; hostile PRs (planted credential, BUSL dependency, critical CVE) refused by gate; a deployed test cluster that must measure green.
- Platform: Rust std-only deterministic matching engine (zero third-party crates, enforced by a build-failing test); 21 Go services on Kubernetes with Argo CD GitOps; PostgreSQL 18 on CloudNativePG, NATS JetStream, Temporal workflows, Valkey, SeaweedFS WORM audit store; Keycloak, OpenFGA and OPA for identity and four-eyes control; Envoy Gateway and Kong at the edge; quickfixgo FIX 5.0 SP2; OpenBao secrets; OpenTelemetry, VictoriaMetrics, Grafana; cosign, Syft, Trivy and gitleaks on every build; open source throughout at \$0 runtime licence.
- CTO Deliverables: 12-month roadmap and 90-day plan; build-vs-buy against incumbent exchange, clearing and surveillance vendors; organisation, hiring and build-and-run operating model; AI-native organisation design; regulatory technology policy pack; 33k-word, 29-diagram investor architecture pack.

Wagey.gg // Founder, Sole Architect & Agentic Engineer (*personal/learning project*)

2025 – 2026 www.wagey.gg

Wagey.gg is a job-discovery and application-assistance platform: multi-source job ingestion, CV-

based matching, browser-extension form filling across Greenhouse, Lever and Ashby, application tracking and Stripe subscription tiers (Free, Pro, Elite); resume tailoring and heuristic alignment scoring exist as gated prototypes. My first dark factory (Shapiro Level 5 — I did not read a line of the code): 14–15 concurrent Claude Code sessions run by hand, with me owning architecture, workstream decomposition, interface contracts, acceptance criteria, integration and release.

- Platform evolution: TypeScript throughout — Next.js/React web app, ingestion services and a Chrome extension; the client moved from an Electron prototype to the extension, the data layer from SQL Server to PostgreSQL, and hosting from Azure App Service to a VPS (PM2, Caddy, GitHub Actions).
- Build pace: circa 178k raw lines of TypeScript and 511 commits at the 17 Feb 2026 snapshot, seven weeks from the first commit.
- LLM routing: task- and context-tiered inference — Groq Llama 3.3 70B for form parsing and low-tier answers, Claude Haiku 4.5 for full-context answers, Ollama for local extraction without per-request fees — with Groq-to-OpenAI fallback for parsing and usage tracking; on an internal form-parsing benchmark Groq matched GPT-4o's extracted field names 96.9% of the time at roughly a tenth of the cost.
- Semantic matching and extraction: four pretrained embedding models (all-MiniLM-L6-v2, jina-embeddings-v2 base-code and base-en, bge-small-en-v1.5) via Transformers.js/ONNX against skill, role and industry taxonomies; exemplar-based region classification with confidence and margin gates; semantic filtering of boilerplate; stemmed token-proximity matching; layered section extraction with semantic and rule-based fallbacks.
- Ingestion and form automation: 35 scraper modules for server-side job ingestion (Playwright, with anti-detection where sources require it) and LLM-driven form parsing, with field verification and integration-test tooling; form filling runs in the user's own browser through the extension.

ACX (AirCarbon) // Chief Technology Officer

Aug 2019 – 2026 <https://acx.net/>

Architected and engineered the world's first fully digital, regulated carbon exchange, featuring a custom-built spot trading engine and RFQ/Auction markets. I personally developed the CertiK-audited Solidity core while building scaling a super lean engineering team to support operations across 5 jurisdictions.

We outcompeted Nasdaq with IDX (national stock exchange Indonesia) selected our stack over theirs, and the business pivoted from a standalone carbon exchange to a multi-tenant network of connected exchanges, with tech being their value and with multiple world class businesses (e.g. EON and B3, national exchange of Brazil) white-labelling our tech.

- Hybrid Split-Ledger Architecture: Designed a novel "Split-Ledger" system decoupling high-frequency centralized matching from decentralized settlement. Implemented a hybrid-permission model combining exchange-controlled execution with ERC-20/1404 on-chain semantics.
- Smart Contract Engineering: Authored the protocol for On-Chain Atomic Spot Trading (token-collateral execution) and Derivatives Position Management (futures, cash settlement, and liquidation engines). Implemented cryptographically verifiable upgrade patterns (hashed code/data) to ensure long-term protocol immutability.
- Engineering Leadership & Delivery: Built the distributed engineering team from Singapore inception to a multi-region regulated footprint (UAE, UK, US, Canada). Delivered a

technology stack explicitly cited by Series A investor Deutsche Börse as the key driver for their investment.

- Cloud-Native Infrastructure: Deployed full-stack AWS ECS Fargate clusters with Cognito identity management. Built high-throughput REST/WebSocket APIs for liquidity providers, ensuring high availability under ADGM (Abu Dhabi) and MAS (Singapore) regulatory TRM guidelines.
- DeFi & Cross-Chain Strategy: Executed the production migration of DLT systems from Binance Smart Chain to Polygon. Engineered DeFi composability via Aave & USDC lending markets integrated directly into the core exchange.
- Security Token Standards (SDAX): Served as Interim CTO for Singapore Digital Assets Exchange (SDAX). Implemented ERC-1400 standards for tokenized equities and built on-chain settlement logic using Chainlink Oracles for real-time fiat-to-crypto pricing, and sat in front of MAS (Singapore regulator) to get them to their in-principal approval.

Maxi.meme // Rust Engineer & Vibemaster (*personal/learning project*)

2025 www.d0m1.com/maxi

For learning and as an experiment, I vibe coded a full DeFi platform with an untrained but highly motivated co-founder; I wanted to see how far the generative tooling could be pushed.

We delivered a fully functional Solana DeFi protocol that conducts Dutch auctions, boots liquidity into Raydium AMM/CLMM pools: on-chain Rust programs, TypeScript micro-services and a React/Vite front-end. I experienced first-hand the limitations of even the current bleeding edge AI tools, especially when combined with untrained non-engineers driving them.

- 2500 + LOC Rust/Anchor program suite with extensive test coverage, modular instruction handlers, comprehensive event stream, secure PDA authority model, and on-chain dynamic bid book that grows via account reallocation.
- Implemented time-decay auction engine (linear/exponential), automatic price discovery, controlled over-liquidity bootstrap using several different techniques, fee-split logic (DAO/admin) and one-click Raydium CLMM pool creation & automatic liquidity migration.
- Hardened security: consistent Rust overflow guards, strict signer checks, compute-unit optimization via batched instructions.
- Built a 5k LOC TypeScript/Express micro-service API (controller-service-repository) with Anchor SDK, Raydium v2 integration, multi-wallet linkage, Twitter (X) OAuth, airdrop orchestration, SQL-Server pooling, Prometheus metrics, Winston logging and Telegram uptime alerts.
- Added performance levers: IP-based rate limiting, layered caching for price feeds, profiles, X data, health-check endpoints, and automatic fail-over keypair rotation.
- Delivered a 7.5k LOC React 18 / Vite / Tailwind frontend: WebSocket live auction feed, 10 + Solana wallet adapters, Lightweight-Charts visualization, multi-hop swap router with slippage guard, social trading cards, mobile-first responsive dashboard.
- Instituted CI/CD & QA: Extensive test-driven Anchor unit test suites: Mocha/Chai integration tests, concurrent-bid stress harness, extensive logical and lifecycle tests; delivered using GitHub Actions pipelines feeding Azure deployments.

Total delivered codebase was ~15k LOC across three repos: production-ready with institutional-grade security and based on core DeFi primitives.

Scoop Wallet // Founder & Maintainer (*personal/learning project*)

2018 – 2020 www.d0m1.com/scpx

Sole developer and maintainer of Scoop wallet; an open-source, pure-JS, multi-device and extendible cryptocurrency wallet, featuring:

- React-based cross platform progress web app (PWA) crypto wallet: ECMAScript 6
- Pure JS CLI/nodejs core wallet, with RPC API
- Private chain EOS backend, with C++ governor smart contract
Multi-asset indexer, fork of Trezor Blockbook: Golang
- Bitcoin protection capability via output and spend of P2SH(P2WPKH)-wrapped non-standard redeem scripts: 1 of 2 multi-sig (“dual sig”) with time lock (OP_CHECKLOCKTIMEVERIFY):
- <https://github.com/Scoop-Tech/scpx-wallet/blob/master/actions/wallet-btc-p2sh.js>

I was the sole developer maintainer of Scoop Wallet; its development has included extensive R&D on forks and multi-node private chain setups of nodeos (EOS blockchain) and Geth (go-ethereum, Ethereum blockchain). I also operated two mainnet Cardano staking pools (tickers [SABAI](#) and [CHAIT](#)) with delegated funds 65m ADA (~\$150mm USD TVL) at peak.

ayondo // Head of Innovation

2016 – 2021

As part of the buyout of my startup (TradeHero, see below), I ran the Singapore R&D engineering team for the acquiring company, and led the technical work to evolve the platform from virtual simulation to a real-money regulated (FCA & BaFin) platform:

- Simulation-to-Execution Engine: Engineered the C# bridge connecting TradeHero’s high-frequency simulation engine with Ayondo’s live banking infrastructure. Re-architected the mobile frontend to interface directly with FIX gateways and Order Management APIs, enabling seamless transition from virtual to real-money trading.
- Identity & Payments Stack: Built a unified KYC/AML onboarding pipeline (C# + native apps) combining mobile biometric verification with backend compliance data stores. Developed secure Payment Processing microservices handling real-time fund deposits, withdrawals, and ledger reconciliation.
- IPO Technical Execution: Led technical due diligence and represented the firm to the regulator in their Reverse Takeover (RTO) and subsequent IPO on the Singapore Stock Exchange (SGX); authored the technical innovation roadmap presented to sponsors and detailed in the public prospectus.

TradeHero // Founder & Chief Technology Officer

2012 – 2016 www.d0m1.com/tradehero

TradeHero was a trading simulation engine on Android & iOS, aggregating real-time feeds from 28 global exchanges. I founded, built, and scaled up a distributed infrastructure to sustain 100k+ DAU and 6m+ users with sub-second latency. Acquired by ayondo in 2016.

- High-Volume Data Engineering: Engineered a zero-downtime live migration of 300 million+

trade records from relational SQL to Azure Table Storage (NoSQL) to resolve write-throughput bottlenecks.

- Distributed Systems: Built a localized, sharded backend (China vs. Global) using C# WebAPI, Redis for hot-path caching, and HDInsight (Hadoop) for massive parallel data processing.
- Technical Due Diligence: Architecture scalability secured a USD \$10m Series A led by Kleiner Perkins (KPCB).
- Sole owner (engineering, devops, DBA) of the C#, TSQL & Azure tiers.
- Sole owner of the product experience, for the first two years.

Fully cloud-based deployment mode: SQL Server, Azure Table Storage (NoSql-based non-relational store), C# WebAPI, Redis, HDInsight Hortonworks Hadoop), serving heterogeneous clients: iOS Objective C & Swift, Android and web.

- Grew technical team from 0 to 35+ staff in Shanghai (native/redeveloped China mobile product) and Singapore (multi-language localized Rest of World (ROW) mobile product) - concept-stage to 6m+ registered installs across 95+ countries: 535% YTD growth 2014-15.
- Designed and built core product to achieve ~40/20/10% at 1, 7 and 30 day cohort retentions – considered to be world-class retentions for mobile gaming sector.
- Built out data centers in mainland China and HK, supporting peak ~100k Daily Active Users (DAU), and 900k MAU; managed in-place live migration of 300m+ row tables to non-relational stores.
- Built WeChat and Facebook viral campaigns with cost reductions in user acquisition (CPA) falling from US\$2+ to < \$0.25.
- Championed at board-level, then designed, delivered and sold live-trading and account acquisition interface for select broker-partners.
- Executed fundraising and private placements across seed stage, series A and series B.
- Filed patents including national-phase WIPO filings in China and US.
- Developed key business relationships through MOU, SOW and implementation phases, bringing in US\$3.5m revenue over 2014-15;
 - Stock Exchange of Singapore (SGX) – multiple sponsorships and competitions for Singapore and other APAC users, relationship spanning 2+ years;
 - Macquarie Group – repeat multi-year engagements in Singapore, Thailand & Malaysia, centered around retail engagement with structured warrants;
 - Kenanga Investment Bank - #1 Malaysian retail broker; pitched and closed to their Retail Equities CEO TH LIVE.
 - Ayondo Markets – advanced stages of joint-venture cooperation including legal consultation and strategy planning on structuring of HK/mainland China vehicle to allow mainland Chinese residents access to overseas CFDs via TH LIVE.
 - Oanda Corporation – competitions, and account acquisition for global FX CFD broker;
 - UBS – white-label offering for UBS staff and clients;
 - Seeking Alpha, Motley Fool – content sharing partnerships;
 - Others including Bursa Malaysia, Stock Exchange of Thailand (SET), Phillip Capital, DBS Vickers, OCBC Securities (Singapore), eToro (global CFDs) Batavia Prosperindo Sekuritas (Indonesia), Qilu Securities & Haitong (China) and CIMB Securities & Maybank Kim Eng (Malaysia).

Allocated and planned the company's USD 2.5m annual technology run rate, drove the search for talent to support growth. I led the hiring from the start: the tech team accumulated a diverse set of talent, from financial markets veterans thru to Apple-hires and artists, interns, and early crypto guys.

The company was selected for inclusion in Microsoft's Beijing-based China Accelerator class of 2014;

we were showcased several times for performance learnings arising from our (I was told, world-first) use of virtualized SQL Server at some non-trivial scale.

Standard Chartered Bank // eCommerce Head of Quality & Release

January 2012 – August 2012

I ran the Quality Assurance and Release Management teams as part of Financial Markets Solutions Delivery (FMSD). I was responsible for the design and implementation of best practices across the release and quality disciplines, as well as for the aggressive hiring and built-out of the group's operational capabilities: 40+ roles triaged & filled within a very short timeframe.

I left this role to found a fintech startup.

Earlier Experience

Citi ~ Global Release Manager, Equities Risk	<i>Oct 2008 – Sept 2011</i>
Microsoft Consulting Services (MCS) ~ SDET	<i>Sept 2005 – Sept 2008</i>
RBS Financial Markets ~ Developer, Currency Options	<i>Feb 2005 – Sept 2005</i>
Deutsche Bank ~ Developer, Equity Derivatives	<i>Oct 2004 – Feb 2005</i>
Barclays Capital ~ Developer Front-office Commodities	<i>Mar 2004 – Oct 2004</i>
Microsoft ~ Software Developer, 3 rd Line Support	<i>Nov 2003 – Nov 2004</i>
Huntleigh Healthcare ~ BA & Senior Developer	<i>Jul 2001 – Nov 2003</i>
X Telecom Ltd. ~ Software Consultant	<i>May 2003 – Jul 2003</i>
Mehdi and Ward ~ Software Architect	<i>Jun 2001 – Jul 2001</i>
InterclubNet ~ Technical Author / Software Developer	<i>Apr 2001 – Jun 2001</i>
Primark (Thomson Financial) ~ Software Developer	<i>Dec 2000 – Apr 2001</i>
IdeaShed ~ Software Architect / Developer	<i>Feb 2000 – Dec 2000</i>
Racal Telecom ~ Software Architect	<i>Mar 1999 – Jul 1999</i>
IT Associates ~ Software Designer / Developer	<i>Feb 1999 – Mar 1999</i>
Northern & Shell ~ Software Designer / Developer	<i>Nov 1998 – Dec 1998</i>
Kimberly-Clark ~ Software Developer	<i>Jul 1998 – Oct 1998</i>
Carland ~ Business Analyst / Software Developer	<i>Apr 1998 – Jul 1998</i>
Aspen Field Marketing ~ Software Developer	<i>Dec 1997 – Mar 1998</i>
Wayne Kerr (sic.) Electronics ~ Software Developer	<i>Oct 1997 – Dec 1997</i>
COM DEV (Europe) ~ Technical Author	<i>Jun 1997 – Sep 1997</i>

Technology Summary

Agentic Engineering & AI (2025–Present)

- Coding-agent workflows: Claude Code (Anthropic), OpenAI Codex CLI, Kimi Code, Grok Build, pi, Cursor.
- LLM integration and inference routing (product side): Groq, OpenAI, Anthropic and Ollama; task- and context-tiered routing with fallback and usage tracking.
- Prompt and context engineering; agentic workflow design.
- Harness engineering: agent constitutions, phase prompts, policy-as-code hooks, bespoke CI gates (77), red-proofed acceptance tests, cold-verifier rounds, truth-gated status sites; control-systems governance of agent fleets (setpoints, sensors, anti-windup, supervisory rulings).

- Multi-vendor agent fleets: 5 harnesses (Claude Code, OpenAI Codex CLI, Kimi Code, Grok Build, pi), 4 providers, 3 capability tiers; orchestrator succession across vendors under one constitution.

NLP, Machine Learning & Embeddings (2026–Present)

- Hugging Face Transformers.js / ONNX: four pretrained embedding models (all-MiniLM-L6-v2, jina-embeddings-v2-base-code, jina-embeddings-v2-base-en, bge-small-en-v1.5).
- Taxonomy matching (skills, roles, industries), exemplar-based classification, semantic filtering, exemplar-based region classification (cosine similarity, margin-gated thresholds).
- NER (Compromise NLP), neural language detection (CLD3/WASM), stemmed token-proximity matching.
- Local inference (Ollama) with pattern-based fallbacks.

Exchange Platform Stack (2026, agent-built under my direction)

- Rust (deterministic matching engine, 65k lines, zero dependencies) · Go (21 services, 106k lines) · Kubernetes (RKE2), Argo CD GitOps · PostgreSQL 18 / CloudNativePG · NATS JetStream · Temporal · Keycloak, OpenFGA, OPA/Rego · OpenBao, External Secrets · Envoy Gateway, Kong · FIX 5.0 SP2 (quickfixgo) · OpenAPI (46 operations), protobuf (6 services / 36 RPCs) · cosign, Syft, Trivy, gitleaks.

Systems & Core Languages (1992–Present)

- C/C++: 30+ years (8086 through Win32; banking/derivatives 2004–2011)
- TypeScript / Node.js: 7+ years (Wagey.gg full-stack web, Chrome extension and automation services; CLI tooling)
- C# & .NET: 20+ years (.NET 1.1 at Microsoft through TradeHero/Ayondo 2016)
- .NET CLR & Internals: 15+ years (reflection, remoting, unsafe context in banking)
- Rust: Solana/Anchor (Maxi.meme, 2,500 LoC hand-built) and a 65k-line deterministic matching engine (2026, agent-built under my direction)
- Python: basic (scripting, data analysis)

Blockchain, Web3 & Cryptography (2017–Present)

- Smart Contracts (Solidity/EVM): 7+ years (Scoop/Ayondo ICO 2017 through AirCarbon)
- Blockchain Infrastructure (Geth/Nodeos/Blockbook/cardano-node): 7+ years (running nodes, migrating chains)
- Web3.js / Ethers.js / DApp Integration: 7+ years (Ayondo TGE through AirCarbon/SDAX)
- Rust & Solana (Anchor): 1+ year (Maxi.meme 2025)
- Golang: 3+ years (Scoop Tech multi-asset indexer/Blockbook fork; 21-service exchange platform 2026, agent-built under my direction)

Cloud & Distributed Architecture (2004–Present)

- Microsoft Azure: 13+ years (TradeHero 2012; Wagey.gg App Service, migrated to a VPS in 2026)
- AWS: 6+ years (AirCarbon: ECS Fargate, Cognito, Lambda)
- Stripe: billing, subscriptions, webhooks
- Distributed Middleware (TIBCO/JMS/Talarian): 8+ years (banking 2004–2012)

Data Engineering & Storage (2000–Present)

- SQL Server / T-SQL: 20+ years (TradeHero, Citi, RBS, Microsoft; Wagey.gg until its PostgreSQL migration); PostgreSQL since 2026 (Wagey.gg, Heimdall Global)
- NoSQL (Redis / Azure Table Storage): 10+ years (300m+ row migration, TradeHero)

- Vector embeddings storage & retrieval (384/768-dim, JSON in SQL, ONNX inference)
- Big Data (Hadoop/HDInsight): 4+ years (TradeHero)

Frontend, Desktop & Mobile (1995–Present)

- Next.js / React (App Router, SSR): 1+ year (Wagey.gg, now Next.js 16 / React 19)
- React / TypeScript / Vite: 7+ years (Scoop PWA 2018; Maxi.meme; Wagey.gg)
- Chrome extension (Wagey.gg form filling); earlier Electron prototype with Playwright automation
- Mobile (iOS/Android): 9+ years (TradeHero & Ayondo 2012–2021)
- WinForms / DirectX 9: 8+ years (banking visualization, vol surfaces)

Testing, Automation & DevOps (1997–Present)

- Playwright: multi-source job ingestion (35 scraper modules) and ATS form automation with field verification and integration-test tooling
- CI/CD: GitHub Actions, Azure DevOps, esbuild, fully bespoke pipelines
- Test frameworks: NUnit, MSTest, Mocha/Chai, test-domain modelling
- Microsoft 3D security practices, static analysis, automated regression (banking)

Education & Interests

- Conditional offer Kings College, Cambridge (Natural Sciences/Philosophy)
- GCSE: 7 As, 3 Bs / A Level: A, B, C
- Hobbies: Fitness, history of ideas, sci-fi, chess.

Childhood Experience

Micro-Ed (MS-DOS) software for schools (own product)

Summer 1992: began writing MS-DOS version in Turbo C (Borland).

Autumn 1993: began marketing MS-DOS version.

Jan 1995: version 5.00 completed with Visual Basic 4.00.

NextBase (now Microsoft)

Summer 1992: one week work experience working with software developers. C programming in a Windows environment.

Electronic Data Systems (EDS)

June 1991: two weeks work experience. C programming for engineering applications, including C programming for PCX graphic file viewer.

Computer Shopper

Nov 1990 and Jan 1991: prizes won in Programmers' Challenge (age 13), written in C.

Personal Computer World

Dec 1990: program published, written in C.

Computer Park

Residential holiday for computer enthusiasts (1990-93) - Met and worked with programmers from NextBase (now Microsoft.)